「Def. Let $A \in M_{m\times n}(F)$ be a m by n matrix

The first elementary Row operation is: for any transposition $(i\ j) \in S_m$,

$$E^{1st}_{i,j} \overset{def}{=} \begin{bmatrix} e_{(i\ j)(1)} \\ e_{(i\ j)(2)} \\ \vdots \\ e_{(i\ j)(m)} \end{bmatrix}$$

($E^{1st}_{i,j} A$ is a i'th and j'th row-changed matrix.)

The second elementary Row operation is: for any non-zero Scalar $c \in F$, and $k \in \{1, \cdots, m\}$

$$E^{2nd}_{c} \overset{def}{=} \begin{bmatrix} e_1 \\ \vdots \\ c \cdot e_k \\ \vdots \\ e_m \end{bmatrix}$$

($E^{2nd}_{c} \cdot A$ is multifying c to k'th row)

The third elementary Row operation is: for any Scalar $c \in F$ and $1 \le i \ne j \le m$,

$$E^{3rd}_{c,i,j} \overset{def}{=} \begin{bmatrix} e_1 \\ \vdots \\ c \cdot e_i + e_j \\ \vdots \\ e_m \end{bmatrix} \leftarrow j\text{th column.}$$

($E^{3rd}_{c,i,j} \cdot A$ is $c \cdot$ (i'th row) adding to j'th row.)

Moreover, these are all invertible,

$$\left(E^{1st}_{i,j}\right)^{-1} = E^{1st}_{i,j}, \quad \left(E^{2nd}_{c}\right)^{-1} = E^{2nd}_{c^{-1}}, \quad \left(E^{3rd}_{c,i,j}\right)^{-1} = E^{3rd}_{-c,i,j}$$

Rank of a Matrix.

Def. Given a Matrix $A \in M_{m\times n}(F)$, define

$$\operatorname{rank} A \overset{\text{def}}{=} \dim \operatorname{Im} L_A.$$

Clearly, $\operatorname{rank} A \le m$.

Prop. Let $A \in M_{m\times n}$, and $P \in M_{m\times m}$ and $Q \in M_{n\times n}$ be invertibles, Then,

$$\operatorname{rank} PAQ = \operatorname{rank} A.$$

Proof. Observe that

$$\operatorname{Im} L_{PAQ} = \operatorname{Im}(L_P \circ L_A \circ L_Q) = (L_P \circ L_A \circ L_Q)[F^n] \underset{L_Q \text{ onto}}{=} (L_P \circ L_A)[F^n]$$

Moreover, since L_P is an Isomorphism $L_P[L_A[F^n]] \cong L_A[F^n] = \operatorname{Im} L_A$. ∎

Thm. Given a Matrix $A \in M_{m\times n}$,

$\operatorname{rank} A = \dim \operatorname{span}\{a_1, \cdots, a_n\}$ where a_i is the i'th Column of A.
(or $a_i = Ae_i$)

Proof. $\operatorname{rank} A = \operatorname{rank} L_A = \dim \operatorname{Im} L_A = \dim \operatorname{span}\{L_A(e_1), \cdots, L_A(e_n)\} = \dim \operatorname{span}\{a_1, \cdots, a_n\}$.

Thm. Given a Matrix $A \in M_{m\times n}$ of $\operatorname{rank} A = r$, then there are invertibles $B \in M_{m\times m}$, $C \in M_{n\times n}$,

such that $BAC = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$

Thm. Let $T: V \to W$ and $U: W \to Z$ be linear maps on fin. dim V, W, Z. Then,

$\operatorname{rank} U \circ T \le \operatorname{rank} U$ and $\operatorname{rank} U \circ T \le \operatorname{rank} T$.

Proof. $\operatorname{Im} U \circ T = (U \circ T)[V] \subseteq U[W] = \operatorname{Im} U$, so $\operatorname{rank}(U \circ T) \le \operatorname{rank} U$.
So, $\operatorname{rank}(U \circ T) = \operatorname{rank}(T^t \circ U^t) \le \operatorname{rank} T^t = \operatorname{rank} T$ ∎

Def. Given $A \in M_{m\times n}(F)$, Consider the linear map

$$L_A : F^n \to F^m ; X \mapsto AX$$

Then, $\ker L_A$ is the solution set of the homogeneous linear system $AX=0$.

Thm. For any kind elementary Row operation E,

$$\ker L_A = \ker L_{EA}.$$

Proof. $AX=0$, then clearly $EAX = E0 = 0$. So $\ker L_A \subseteq \ker L_{EA}$.
Conversely, since E is invertible, $EAX=0 \Rightarrow AX = E^{-1}0 = 0$.

Thm. Given a m by n matrix $A \in M_{m\times n}(F)$ and $B \in F^m$,
if $s \in F^n$ satisfies $As = B$, then

$$x \in F^n \text{ satisfies } Ax = B \iff x \in s + \ker L_A.$$

Proof. If $Ax = B$, then $Ax = B = As$, so $A(x-s)=0$. That is, $x - s \in \ker L_A$, or $x \in s + \ker L_A$
Conversely, $x \in s + \ker A$ be given. That is, $x = s + y$ for some $y \in \ker L_A$, so

$$Ax = A(s+y) = B + 0 = B.$$

Corollary. $Ax=b$ has only one solution iff A is invertible.
Corollary. Given linear System $Ax=b$ where $A \in M_{m\times n}$, $x \in F^n$, $b \in F^m$,

$Ax=b$ has a solution if and only if $\operatorname{rank} A = \operatorname{rank}(A|b)$.

Proof. Since $Ax=b$ for some $x \in F^n$ if and only if $b \in \operatorname{Im} A$, and

$$\operatorname{Im} A = \operatorname{Span}\{Ae_1, \cdots, Ae_n\}$$

So, Proved. $\boxtimes$

┌ In Inner Product Space, we can use inner product to find "best" approximation solution.

┌ Thm. Let $A \in M_{m\times n}(\mathbb{C})$ and $y \in \mathbb{C}^m$ be given.
Then, there exists $x_0 \in \mathbb{C}^n$ such that

$$A^*Ax_0 = A^*y \quad \text{and} \quad \|Ax_0 - y\| \leq \|Ax - y\| \text{ for all } x \in \mathbb{C}^n.$$

In particular, if rank $A = n$, then $x_0 = (A^*A)^{-1} \cdot A^*y$.

Proof. Given $y \in \mathbb{C}^m$, there exists an orthogonal Projection Ax_0 on $\operatorname{Im} A$.
That is, $\langle Ax_0 - y, Ax\rangle = 0$ for all $x \in \mathbb{C}^n$.
And, since $\langle Ax_0 - y, Ax\rangle = \langle A^*Ax_0 - A^*y, x\rangle$, so $A^*Ax_0 - A^*y = 0$.
(Specifically, for orthonormal basis $\{\alpha_1, \cdots, \alpha_r\}$ for $\operatorname{Im} A \subseteq \mathbb{C}^m$,

$$Ax_0 = \sum_{i=1}^{r} \langle y, \alpha_i\rangle \cdot \alpha_i.)$$

Moreover, if rank $A = n$, then rank $A^*A = n$, so invertible.